AERIS - Aircrew Explainable Readiness & Intervention System

1. Description of Proposed Phase I Technical Effort

Naval Aircrew readiness is multidimensional and time-varying. Physiological state, visual attention, fatigue, cognitive workload, perceived readiness, and interaction with aircraft systems can change before those changes become visible as procedural errors or degraded decision-making. The Navy identifies a current gap in dynamic, predictive, personalized readiness assessment and specifically seeks an intelligent human-performance system that can monitor and predict cognitive, physical, and psychological vulnerabilities while augmenting aeromedical officers with actionable information.[1] Groundbreaker Solutions LLC proposes AERIS, a sensor-agnostic human-performance intelligence architecture designed to forecast emerging vulnerability, preserve the evidence behind each forecast, and keep the aeromedical officer - not the algorithm - as the decision authority.

AERIS centers on a continuously updated Human-Performance Digital Twin: a person-specific temporal state representation that combines population priors with an individual's expected range under comparable task demands. Rather than treating one heart-rate threshold, gaze metric, or self-report score as readiness, AERIS fuses multimodal evidence, models data quality and missingness, estimates uncertainty, and compares current observations with longitudinal baseline. The approach is deliberately conservative about inference: simulator event logs directly measure procedural error and decision latency; physiological and ocular signals provide evidence associated with latent constructs such as workload, fatigue, attentional narrowing, stress response, and spatial-orientation vulnerability. AERIS does not claim that a wearable independently diagnoses spatial disorientation, burnout, or any medical condition.

The technical innovation is the integration of four capabilities into one aeromedical workflow: (1) individualized longitudinal state estimation instead of population-only thresholds; (2) uncertainty-aware multimodal fusion that treats signal quality as model evidence; (3) explainable vulnerability forecasting that links predictions to source data, baseline departure, provenance, and limitations; and (4) closed-loop readiness optimization in which an expert-selected training or prehabilitation action is followed by observed outcome data that update the individual model. An exploratory crew-readiness extension will test whether crew interaction and mission context add useful information beyond individual-only models, without making crew-assignment or fitness-for-duty decisions.

The MRS-15 provides a particularly strong Navy-specific anchor. Taylor et al. developed and validated the 15-item Military Readiness Scale in 164 U.S. Naval Aircrew, yielding Total, General-Psychological, Technical, and Physical readiness measures and identifying prospective operational validation as a logical next step.[2] AERIS therefore treats MRS-15 as a validated perceived-readiness input and longitudinal context measure - not as ground truth - and tests whether combining it with objective performance, physiological, ocular, voice-derived, and mission-context evidence improves prospective vulnerability estimation.

2. Phase I Technical Objectives

Objective	Feasibility question / Phase I evidence
O1 - Aeromedical CONOPS	Can readiness evidence and user roles be represented coherently? Deliver CONOPS, ontology, and metric traceability.
O2 - COTS multimodal data plane	Can heterogeneous physiological, ocular, and simulator streams be synchronized, quality-tagged, and replayed through a common model?
O3 - Individualized forecasting	Can a person-specific temporal model detect and forecast scripted workload/tunneling/orientation-risk states with calibrated uncertainty?
O4 - Explainable decision support	Can every warning be traced to source evidence, baseline departure, data quality, confidence, and expert-reviewable rationale?
O5 - Phase II readiness	Can the concept be converted into a statistically rigorous, IRB/DoN HRPO-ready validation and transition plan?

Phase I thesis: prove that heterogeneous readiness evidence can be acquired or replayed, synchronized, quality-controlled, personalized, fused, forecast, explained, and presented for aeromedical action - without making Phase I success dependent on human-subject experimentation.

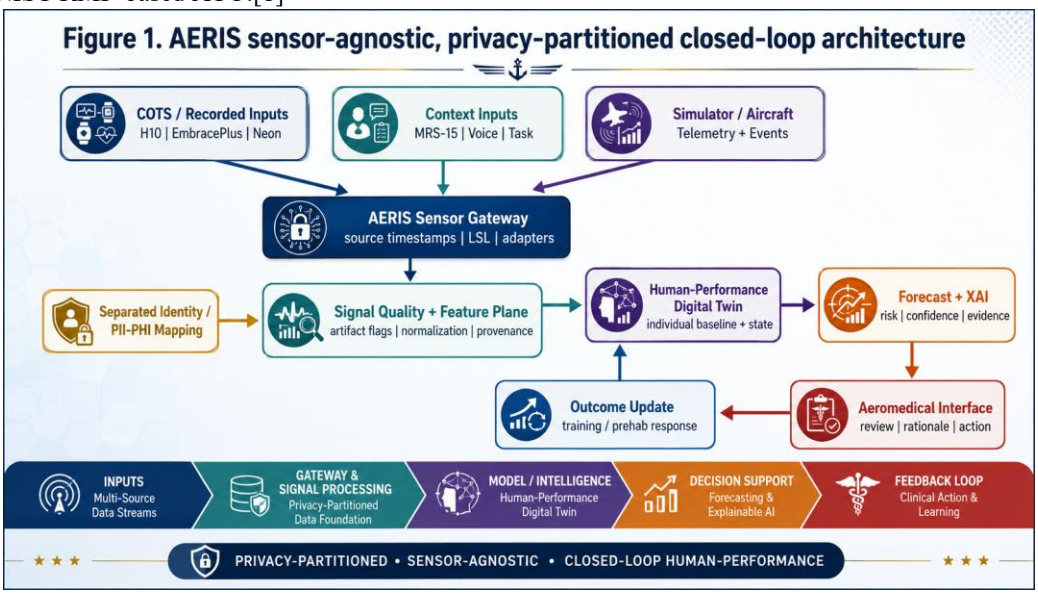
3. Phase I Statement of Work - Base Period (Months 1-6)

3.1 CONOPS and Preliminary System Architecture

AERIS is organized around three actors: the aircrew member, the aeromedical officer, and the training/human-performance environment. The aircrew member generates task-performance, physiological, ocular, and contextual evidence. The training environment supplies task demand, aircraft/simulator state, automation events, and objective performance outcomes. The aeromedical officer receives a role-appropriate, evidence-first view of readiness trends and forecasted vulnerability. AERIS may identify patterns that warrant review and may present SME-approved categories of training, recovery, conditioning, or further assessment; it will not independently diagnose, prescribe clinical treatment, or determine medical qualification.

The architecture intentionally separates sensing from intelligence. Phase I will use COTS devices as representative data sources, but each source is isolated behind an adapter and canonical event/time-series contract so Phase II can substitute flight-compatible or Government-preferred hardware. Source-native timestamps are retained and mapped into a common research time base using Lab Streaming Layer (LSL) where practical. LSL is well suited to multimodal experiments because it supports synchronized collection and clock correction across heterogeneous streams; Pupil Labs Neon exposes native LSL integration.[10] Vendor-cloud dependencies, such as Empatica data access, remain outside the core analytics boundary and can be converted into timestamp-preserving replay streams.

The edge data plane performs synchronization, quality assessment, artifact/missingness tagging, privacy-minimizing feature extraction, and provenance capture before data enter the digital-twin model. Raw scene video or voice need not be retained when derived gaze, fixation, pupil, head-motion, speech-rate, pause, or prosodic features are sufficient. A separate identity mapping domain keeps direct identifiers apart from pseudonymous analytical records. This anticipates the solicitation's requirement for strict protection and de-identification of PII/PHI and the Phase III path to a NIST RMF-based ATO.[1]



The Human-Performance Digital Twin maintains a vector rather than a single score: General/Psychological readiness, Physical readiness, Technical readiness, cognitive workload, fatigue, attention stability, stress response, spatial-orientation vulnerability, automation-trust context, task performance, data quality, and uncertainty. A population model supplies a cold-start prior; repeated person-specific observations adapt expected ranges. Uncertainty expands when a sensor becomes unreliable, a modality disappears, or the operating context lies outside the modeled range. The core hypothesis is not that a complex model is inherently better; Phase I will test whether individualized multimodal forecasting materially improves over interpretable population baselines while preserving transparency.

3.2 Base Task 1 - Aeromedical CONOPS, Requirements, and Readiness Ontology

Groundbreaker will map the solicitation's cognitive, physical, psychological, trust, and human-system concerns into measurable constructs and explicitly define which variables are direct outcomes versus inference features. The task will produce the user-role matrix, readiness ontology, metric-to-source matrix, interface requirements, privacy boundary, and requirement-to-test traceability matrix. Aeromedical and psychophysiology SMEs will review terminology, acceptable interpretation, intervention boundaries, and the design of the Phase II validation plan.

3.3 Base Task 2 - COTS Sensor Gateway, Synchronization, and Data Quality

Phase I will acquire or integrate representative COTS hardware to demonstrate real device interfaces while keeping the analytical architecture hardware independent. The recommended core suite is Polar H10 + Empatica EmbracePlus + Pupil Labs Neon. Polar H10 provides ECG at 130 Hz plus RR/HR and motion through the Polar BLE SDK and has strong independent validation for HR/HRV use under appropriate conditions.[3] EmbracePlus contributes electrodermal activity (EDA), PPG/BVP, skin temperature, and motion, but wrist PPG is explicitly treated as motion-sensitive evidence rather than a cardiac reference.[4] Neon contributes 200 Hz gaze/pupillometry/eye-state data, head-motion/IMU streams, scene context, and direct LSL support, making it especially useful for instrument-scan, fixation, blink, gaze-entropy, and attentional-tunneling features.[5]

Priority	Device / signals	Phase I use and limitation
Core	Polar H10 - ECG, RR/HRV, IMU	Cardiac anchor; direct BLE adapter; chest-strap form factor is a research reference, not an operational mandate.
Core	Empatica EmbracePlus - EDA, PPG, temperature, IMU	Autonomic/thermal/motion evidence; timestamp-preserving cloud/file replay; PPG quality gated by motion.
Core	Pupil Labs Neon - gaze, pupil, blink, eye state, head pose, scene	Direct attention/tunneling evidence; native LSL; research eyewear is not assumed flight-qualified.
Alternate	Shimmer3 GSR+ - EDA, PPG, IMU	Substitute for Empatica when local/raw-stream control is preferred; useful for bench/research integration.
Stretch	Muse 2/S - 4-channel EEG, PPG, IMU	Ablation channel for workload/fatigue research; not required for feasibility and may conflict with helmet/eye-tracker form factor.
Context	Oura / WHOOP / Garmin	Optional prior-night sleep/recovery/strain context through cloud APIs; never a real-time cockpit alert dependency.

AERIS will not assume that every channel is trustworthy at every moment. Each adapter emits data plus a Signal Quality Index (SQI), availability flag, timestamp provenance, and transport diagnostics. ECG processing will flag gaps, implausible beat intervals, and poor morphology; PPG quality will be cross-checked against acceleration and pulse morphology; EDA will be screened for dropout/flatline/abrupt artifacts; gaze will be scored for valid-sample percentage, occlusion, and calibration/offset status; optional EEG will be checked for channel dropout and gross artifacts. These quality variables enter the fusion model so the system can reduce confidence instead of silently converting bad data into a high-confidence readiness warning.

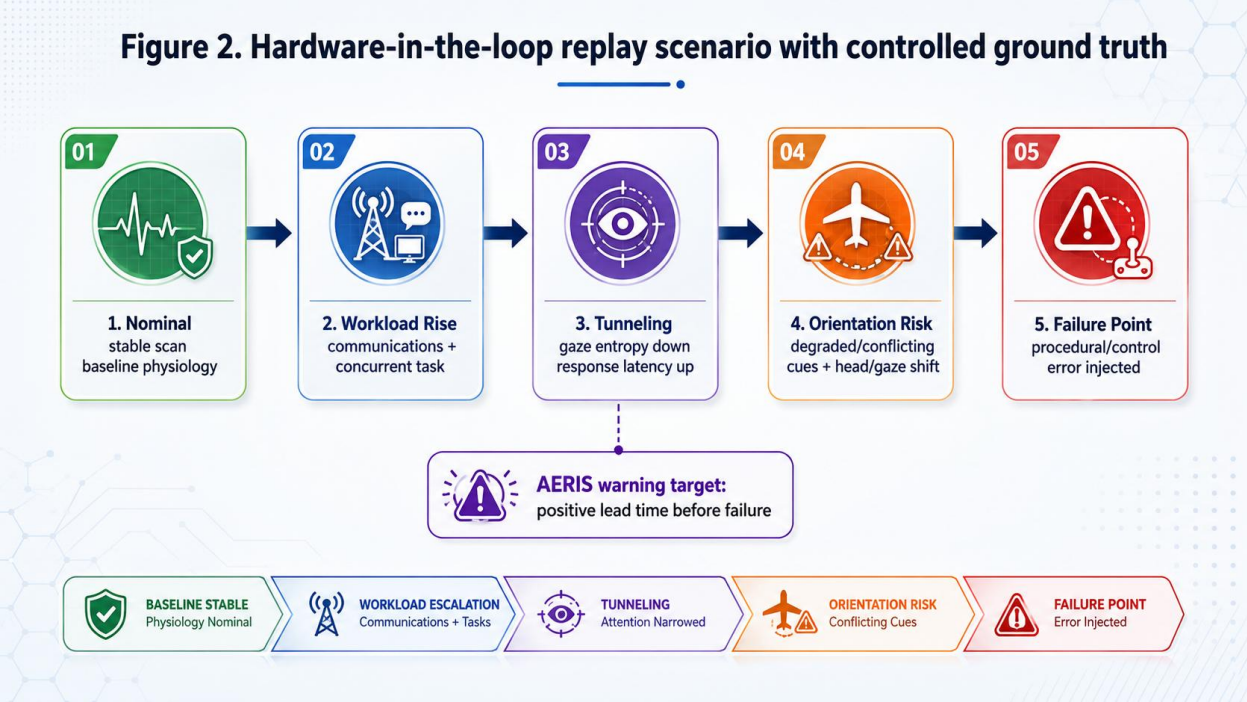
Streams use multiple time scales. Fast ocular/EDA changes are evaluated in short windows, cardiac trends in intermediate windows, and longitudinal readiness or HRV context in longer windows. Person-specific normalization will be represented initially as standardized deviation from comparable baseline (for feature i at time t, $z_{i,t} = [x_{i,t} - \mu_i]/\sigma_i$), with mission/task context conditioning expected ranges. The architecture also retains absolute values so clinically or operationally meaningful limits can be incorporated later under Government/SME direction.

3.4 Base Task 3 - Individual Digital Twin and Multimodal Inference

Groundbreaker will implement two principal benchmark model families: an interpretable logistic/generalized-additive baseline and a nonlinear tree/boosting model with calibrated probabilities. A lightweight state-space or hidden-state layer will carry temporal continuity and personalized baseline adaptation. Phase I will compare population-only, individualized, and multimodal variants; run modality-ablation tests; and inject missing or corrupted streams. Model selection will be based on discrimination, calibration, lead time, robustness, and explanation stability rather than raw accuracy alone. Crew/context modeling is a secondary experiment: the system will test whether interaction and automation/task context provide incremental value over individual-only state, but will not make personnel-assignment recommendations.

3.5 Base Task 4 - Proof-of-Concept Simulation and Hardware-in-the-Loop Replay

The Phase I proof of concept will focus on one Navy-aligned exemplar: progressive workload with attentional narrowing and incipient spatial-orientation vulnerability during a simplified instrument-flight task. The topic itself identifies predicting disorientation from physiological cues as an example Phase I demonstration.[1] AERIS will not attempt to recreate an aircraft trainer. Instead, a deterministic scenario engine will generate aircraft/task state, instrument-area-of-interest events, communications demand, automation changes, procedural prompts, expected responses, and injected flight-control deviation. This produces authoritative event labels and direct measurements for procedural error, response latency, tracking error, and missed cross-checks.



Synthetic physiology/behavior streams will be literature-calibrated and intentionally controllable: ECG/RR trends, EDA response, respiration tendency, pupil response, blink/eye-state, gaze distribution/entropy, head-motion relationship, speech-rate/pause/prosodic features, and session-level MRS-15 context. Existing vendor sample files, public or appropriately licensed de-identified research recordings, and device-native file formats will be used to prove adapter compatibility with realistic signal morphology. Hardware-in-the-loop testing will validate SDK connectivity, packet/file ingestion, timestamp mapping, reconnect behavior, event-marker alignment, and replay. Completion of the Base will not depend on prospectively collecting physiological or psychological research data from people.

The controlled scenario enables a clean causal chain for engineering validation: known workload demand increases; visual scanning narrows; physiological features depart from baseline; response latency and control variability begin to deteriorate; orientation cues become degraded/conflicting; then a predefined procedural/control failure occurs. AERIS succeeds only if it can forecast the scripted vulnerability before the failure while communicating uncertainty and the actual evidence used. Aviation research supports using cardiac/autonomic measures, pupil response, fixation patterns, gaze entropy, and eye metrics as workload/fatigue evidence, while simulator studies show spatial-disorientation events alter visual-attention behavior.[6-9] Those associations motivate feature classes; they are not treated as single-sensor diagnostic rules.

For every forecast the demonstration will answer five reviewer-visible questions: What changed? How certain is AERIS? Why did the model change? What information is missing or unreliable? What may an authorized aeromedical expert consider next? A simulated intervention - for example workload recovery, targeted simulator repetition, or further assessment - will then alter the scripted state trajectory and the subsequent observations will update the twin. That final outcome-update step distinguishes AERIS from a one-way dashboard and makes the proposed closed-loop readiness concept testable.

Human Subjects Research Boundary

The Base is designed so technical success does not require Human Subjects Research (HSR). No recruitment, experimental intervention, or prospective aircrew physiological/psychological data collection is required to meet the Phase I objectives. If any existing human-derived dataset or external design activity could constitute HSR, Groundbreaker will obtain the required determination before use. The Phase I Option will prepare the Phase II human-use protocol, but no human testing will begin without local IRB approval and subsequent DoN Human Research Protection Official approval as required by the topic, 32 CFR Part 219, DoDI 3216.02, 10 U.S.C. 980, and DFARS 252.235-7004.[1]

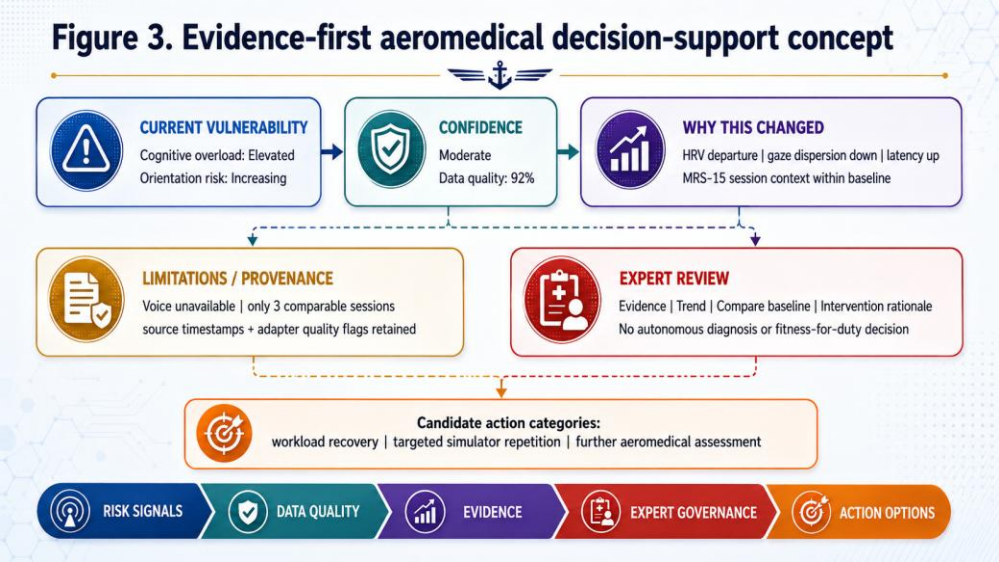
3.6 Base Task 5 - Evaluation, Explainability, and Aeromedical HMI

AERIS evaluation separates engineering feasibility from future operational validity. Synthetic or replayed data can prove integration, synchronization, robustness, state-estimation behavior, calibration, and explanation traceability; they cannot prove that the same model improves Naval Aircrew performance. Phase I therefore establishes the measurement infrastructure and test methods needed to evaluate the Navy's Phase II targets - statistically significant 15-20% reductions in tactical/procedural errors under simulated stress and 10-15% improvement in decision latency without degraded accuracy.[1]

Dimension	Phase I engineering measure	Phase II linkage
Data integrity	>=99% expected samples/events accounted for or explicitly flagged; packet loss and reconnect characterization	Reliable operational sensing
Synchronization	Common event-marker offset/jitter characterized by adapter; target <=20 ms for locally controlled streams	Cross-modal feature validity
State discrimination	AUROC >=0.85 and precision/recall >=0.80 on primary scripted scenarios	Detect relevant vulnerability states
Calibration	Brier score/calibration curve; confidence decreases under missing or degraded streams	Interpretable probability/confidence
Forecasting	Positive warning lead time; false alarms/hour at selected operating point	Proactive intervention before failure
Personalization	Individualized model compared with population-only baseline	Personalized readiness assessment
Fusion/robustness	Modality ablation, dropout, jitter, corrupt-stream challenge	Sensor down-select and graceful degradation
Explainability	100% alerts include sources, direction, baseline comparison, quality, confidence, limitations	Expert review and trust calibration

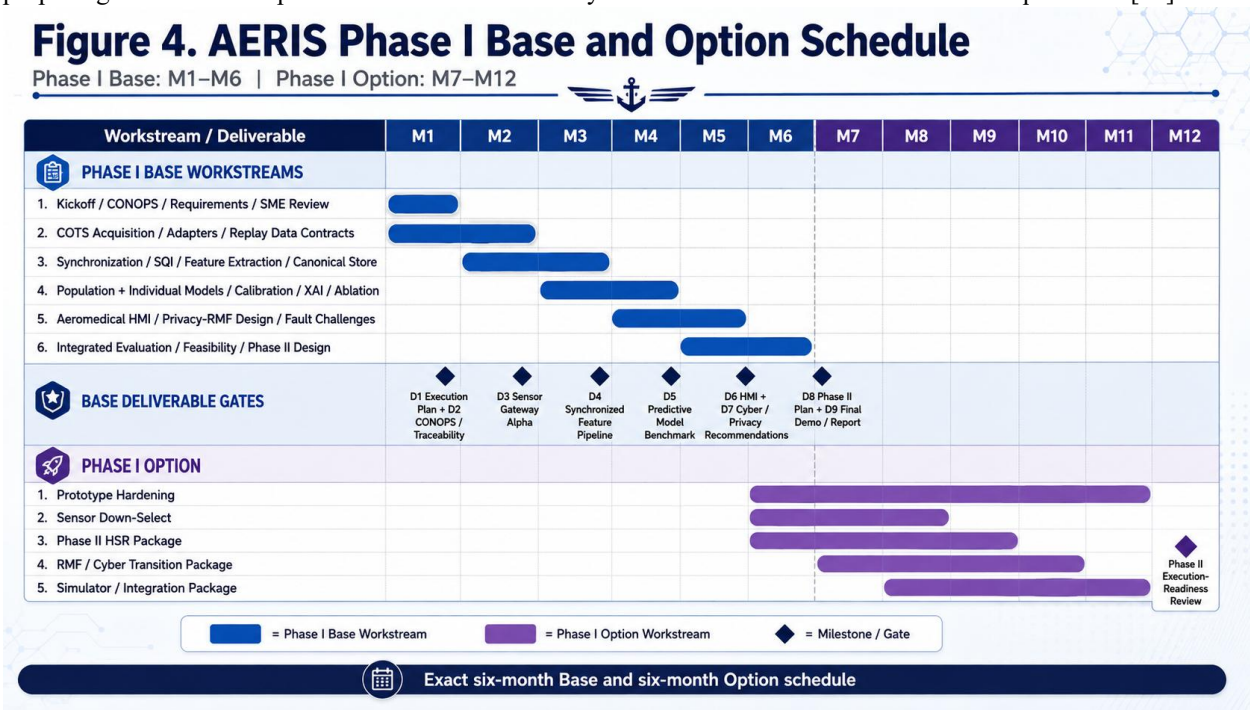
Explainability will be implemented as an evidence ledger, not as a claim of causality. Feature attribution can identify inputs that affected a model prediction; it does not prove that manipulating those inputs changes readiness. AERIS therefore separates three layers: model explanation (which inputs drove the prediction), evidence interpretation (what established physiological or behavioral relationship the pattern is consistent with), and expert decision support (what prehabilitation, training, recovery, or further-assessment categories are within the aeromedical officer's authority). Causal efficacy of interventions remains a Phase II research question requiring an approved prospective design.

The prototype interface will show current vulnerability trends, confidence, source quality, comparison to individual baseline, evidence contributions, unavailable modalities, number of comparable prior observations, and an intervention-rationale view. It will avoid simplistic red/yellow/green medical-clearance labels unless later approved by aeromedical SMEs and the Government. AERIS will also expose human-system context: for example, a workload warning may be linked to an automation transition, narrowed instrument scan, delayed procedural response, and rising control variability rather than appearing as an unexplained 'stress score.' This directly supports the topic's goals for human-system vulnerability identification, trust calibration, and transparent data for aeromedical review.[1]



3.7 Base Task 6 - Integrated Demonstration, Deliverables, and Phase II Plan

Months 5-6 will integrate the sensor gateway, replay framework, feature pipeline, digital twin, predictive models, evidence ledger, and aeromedical interface into a repeatable demonstration package. Groundbreaker will execute scripted workload, tunneling, orientation-risk, fatigue-trend, and sensor-fault scenarios; document results against the engineering gates; record open limitations; and produce a Phase II prototype and validation plan. The company will retain ownership of core architecture, software, modeling, integration, UI, and program management so the proposing small business performs well above the Navy's minimum two-thirds Phase I work requirement.[11]



3.8 Phase I Option - Exactly Six Months (Months 7-12)

If exercised, the Option will bridge directly to Phase II without making progress dependent on premature human testing. O1 will evaluate flight-relevant form factor and operational substitutes for research devices; O2 will expand simulator/aircraft telemetry adapters based on NAVAIR direction; O3 will prepare the Phase II experimental design, safety plan, power-analysis inputs, IRB materials, DoN HRPO package, and statistical analysis plan; O4 will harden missing-stream handling, probability calibration, individual-baseline adaptation, and model APIs; O5 will mature RMF artifacts including system boundary, data-flow diagram, threat model, logging, dependency/SBOM inventory, retention policy, and control traceability; O6 will conduct a Phase II test-readiness review and deliver the refined prototype/integration package.

Cybersecurity, Privacy, and Technical Risk

The topic projects CMMC Level 2 (Self) and anticipates a Phase III ATO under the NIST Risk Management Framework.[1] AERIS will therefore separate identity, clinical/readiness data, and model features; minimize raw sensitive-data movement; use pseudonymous analytical identifiers; apply least-privilege role-based access, encryption, auditable provenance, configurable retention, dependency management, and explicit system boundaries. Phase I will not require a commercial cloud: core analytics can operate locally/on-premises, with vendor-cloud connectors isolated behind adapters.

Risk	Mitigation
Motion artifact in wrist PPG	Use H10 ECG as cardiac anchor; acceleration-aware SQI gates wrist-derived pulse/HRV.
Sensor dropout/clock drift	Retain source timestamps; LSL/event markers; fault injection; confidence degrades with quality.
Research eyewear not flight-compatible	Use Neon as reference sensor only; Phase II down-selects helmet-compatible ocular sensing.
Synthetic performance overclaim	Limit Base claim to system/analytical feasibility; human validity and operational benefit reserved for approved Phase II studies.
Algorithmic opacity	Interpretable baseline + calibrated nonlinear model + evidence ledger + explicit uncertainty.

4. Related Work and State of the Art

AERIS is informed by both an internal architectural lineage and the post-2018 human-performance evidence base. Groundbreaker's 2018 mil.API.fit concept examined COTS biometric sensing, an edge/service-bus architecture, machine-learning correlation of physiology with training performance, psychometric inputs, hierarchical dashboards, black-box explainability concerns, adaptive training, and team/cohesion reasoning. That document is prior design thinking, not claimed funded past performance. AERIS advances the concept by making the aeromedical officer the primary expert user, adding eye tracking and task/automation context, explicitly modeling signal quality and uncertainty, establishing individual longitudinal baselines, preserving evidence provenance, and closing the intervention/outcome loop.

2018 mil.API.fit concept	AERIS advancement
Wearable PPG/EDA/temp/motion	Sensor-agnostic ECG/EDA/PPG/temp/motion + gaze/pupil/head + voice + task telemetry + MRS-15
Edge service bus + military cloud	Edge-first, cloud-enabled architecture with privacy segmentation and offline/degraded operation
ML prediction of training outcomes	Temporal vulnerability forecast with calibration, uncertainty, missing-modality robustness, and ablation
Psychometric instruments	MRS-15 as validated perceived-readiness modality, explicitly not ground truth
Instructor dashboard / training modification	Aeromedical evidence ledger and expert-governed prehabilitation/training rationale
Team/cohesion reasoning	Secondary crew/context model testing incremental value without personnel-selection decisions

Current research makes this modernization technically plausible but also argues for restraint. Polar H10 has been independently validated for RR/HR and many linear HRV measures, supporting its use as a low-cost cardiac reference in research contexts.[3] Wrist PPG devices can provide valuable pulse and autonomic context, but motion materially affects accuracy and HRV quality, which supports AERIS's quality-aware fusion rather than unconditional averaging.[4] Mobile eye tracking has reached sufficient frequency and integration maturity for scan behavior, fixation, pupil, blink, and head-motion research; Pupil Labs Neon provides a practical Phase I implementation while published evaluations caution against overinterpreting fine microsaccadic/event-classification detail.[5]

Aviation-specific literature supports the feature strategy. Fighter-pilot and simulated-flight studies have reported workload-related changes in gaze entropy, fixation/scan behavior, cardiac-autonomic measures, and combinations of HRV and eye metrics.[6-8] Simulator-induced spatial-disorientation research has found measurable ocular-attention changes, supporting gaze and head/aircraft context as risk evidence while underscoring that disorientation is a multimodal operational state rather than a single biosignal.[9] The MRS-15 work adds a Navy-specific subjective readiness construct with General-Psychological, Technical, and Physical dimensions and highlights the need for prospective operational validation.[2]

The innovation is therefore not 'AI on wearables.' AERIS unifies individualized longitudinal state estimation, multimodal quality-aware fusion, calibrated temporal forecasting, evidence/provenance-based explainability, expert-governed action, and outcome updating within a privacy-partitioned architecture tailored to the solicitation's aeromedical workflow. Phase I deliberately benchmarks interpretable and nonlinear models and tests whether each added modality or personalization step provides measurable value. A sensor or model that does not improve performance, calibration, lead time, robustness, or explanation quality can be removed before Phase II.

Groundbreaker will coordinate outside expertise through narrowly scoped aeromedical, psychophysiology/biostatistics, and aviation-human-factors consultation. The proposer will perform the core architecture, software, simulator/replay framework, model integration, testing, HMI, and program management. Consultant outputs will be technical reviews and validation artifacts, not outsourced ownership of the innovation.**5. Relationship with Future Research and Development**

Successful Phase I completion will establish that AERIS can ingest representative COTS and simulated evidence, synchronize and quality-tag streams, maintain an individualized temporal readiness state, forecast scripted vulnerability with calibrated uncertainty, explain the evidence to an aeromedical user, and update the model after a simulated intervention. Phase II will convert that analytical feasibility into an operational prototype with Government-directed sensor and simulator interfaces, approved human-subject validation, statistically powered comparisons, and direct measurement against the topic's tactical/procedural error and decision-latency targets. The Phase I Option will prepare the HSR, data-management, safety, statistical, integration, and RMF artifacts needed to minimize Phase II startup delay.

Phase II human testing will require local IRB approval and subsequent DoN HRPO approval before recruitment/testing, with compliance to the authorities specified in the topic. Sensor down-selection will also address flight/simulator form factor, helmet compatibility, electromagnetic/environmental constraints, data ownership, licensing, and whether an operational device can provide the required raw or derived features. AERIS remains sensor agnostic so the research devices used to establish Phase I feasibility do not become architectural dependencies.

6. Key Personnel

Jason L. Lind - Principal Investigator / Technical Lead. Mr. Lind will own system architecture, integration, digital-twin design, predictive-model integration, explainability, technical risk, and prime deliverables. He has more than two decades of enterprise/software architecture experience spanning .NET, Azure/cloud integration, cybersecurity, real-time/distributed systems, AI-assisted applications, and modeling/simulation. Relevant recent work includes real-time mobile/web architectures, modernization of a U.S. Space Force weather-balloon tracking application, and simulation-oriented decision-support concepts.

Aerospace Medicine / Naval Aircrew Human Performance SME - [TBD before submission]. This consultant will review the CONOPS, readiness/vulnerability taxonomy, screening-versus-diagnosis boundaries, MRS-15 interpretation, intervention categories, human-subject safety strategy, and aeromedical HMI. Desired qualifications include aerospace medicine, aviation physiology, aerospace experimental psychology, military/civil flight medicine, or directly relevant aircrew human-performance experience.

Psychophysiology / Biostatistics SME - [TBD before submission]. This consultant will review signal feature definitions, physiological interpretation, signal-quality rules, individualized/longitudinal modeling, calibration, missing-data strategy, model comparison, Phase II effect-size assumptions, statistical analysis, and power methodology. The role is intentionally scientific-governance oriented so AERIS does not convert correlational wearable features into unsupported causal claims.

Aviation Human Factors SME - [TBD or combined with aeromedical SME]. This consultant will review the simplified flight task, workload manipulations, instrument/AOI logic, automation/trust conditions, procedural-error definitions, decision-latency measures, and crew/context experiment. Groundbreaker will perform at least two-thirds of Phase I work in both Base and Option; final consultant hours and cost allocations will be structured accordingly.[11]

7. Facilities / Equipment; Foreign Citizens; Similar Support

Phase I requires no wet laboratory or hazardous-material facility. Work will use Groundbreaker's controlled software-development environment, secure source/build resources, network test services, and COTS research sensors. Planned equipment includes Polar H10, Empatica EmbracePlus or alternate Shimmer3 GSR+, Pupil Labs Neon, and optional Muse EEG; final purchases/pricing will be justified in Volume 3. Phase I analytical testing will use synthetic/replay data unless an appropriate determination/approval permits otherwise; environmental impacts are limited to ordinary office/electronic equipment use.

Foreign Citizens: none proposed in this draft. Data rights: no restrictions beyond applicable SBIR/STTR rights are asserted in this draft; identify any pre-existing privately developed software or technical data before final submission.

8. Commercialization / Transition Plan Summary

AERIS commercialization is transition-led. The initial Government customer is Naval Aviation's training/human-performance/aeromedical ecosystem: the Phase III topic language contemplates deployment in Naval Aviation training units, aeromedical facilities, mission-rehearsal systems, physiological monitoring infrastructure, and multiple Naval Air Stations, followed by broader military, allied, civilian aviation, and other high-reliability applications.[1] Groundbreaker's product strategy is therefore not to manufacture a proprietary wearable. The commercial asset is the hardware-agnostic intelligence layer, validated sensor/simulator adapters, individualized models, explainable expert workflow, and secure deployment/integration package.

The product can mature as four modules: AERIS Edge for sensor/telemetry synchronization, signal quality, privacy filtering, and local inference; AERIS Twin for individualized longitudinal state and forecast services; AERIS Expert for aeromedical evidence review and intervention/outcome tracking; and AERIS Analytics for appropriately de-identified cohort/training-system analysis. This modular approach supports local/on-premises deployment, Government-approved cloud environments where appropriate, and vendor substitution without retraining the organization around a single commercial wearable.

Phase I commercialization milestones are a demonstrated COTS adapter set, repeatable analytical PoC, aeromedical HMI, data-rights/security architecture, and Phase II integration/validation package. Phase II milestones are a Government-directed operational prototype, approved human validation, simulator/training integration, sensor down-select, measurable performance evaluation, and transition-site demonstration. Phase III would package the validated capability for production integration, RMF/ATO, sustainment, additional adapter development, model monitoring, and support services. The closest dual-use market is civil aviation training and safety - airlines, flight-training organizations, simulator centers, aviation human-factors organizations, and other high-reliability operations where fatigue, workload, automation trust, and human-performance risk are economically significant and where

expert oversight is already part of the workflow. Healthcare/emergency response and elite-performance markets remain secondary until aviation validation is established.

Groundbreaker's defensible differentiation is not access to the same COTS devices available to competitors. It is the validated mapping from heterogeneous evidence to individualized state, uncertainty-aware vulnerability forecast, auditable explanation, expert-governed intervention, and outcome update - combined with a privacy/security architecture suitable for sensitive human-performance data. Phase I proves the evidence can be acquired, synchronized, quality-controlled, fused, predicted, explained, and acted upon; Phase II determines whether those predictions improve real Naval Aircrew outcomes; Phase III turns the validated capability into a deployable readiness product.

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